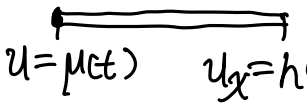
数理方程

第五周

2026032?

非齐次BC（边界条件）
- 本征函数展开法
- BC齐次化（减去法）
  - BC齐次化（不管PDE）
  - BC+PDE同时齐次化

$$u_{tt}-a^2u_{xx}=f(x,t),\quad 0<x<l,\ t>0$$



$$u|_{x=0}=\mu(t),\quad u_x|_{x=l}=h(t),\quad u|_{t=0}=u_t|_{t=0}=0$$

直接分离变量：$u=X(x)T(t)\Rightarrow u|_{x=0}=X(0)T(t)=\mu(t)\Rightarrow T(t)\overset{?}{=}\frac{\mu(t)}{X(0)}$

展开法（移力法）

"任意"$u(x,t)$可作傅里叶展开，$T_n(t)=\frac{1}{\|X_n\|^2}\int_0^l u(x,t)X_n(x)\,dx$

$$u(x,t)=\sum_n T_n(t)X_n(x)=\sum_{n=0}^{\infty}T_n(t)\sin\left(n+\frac{1}{2}\right)\frac{\pi x}{l}$$

$u|_{x=0}=u_x|_{x=l}=0$

$\Rightarrow X_n=\sin\left(n+\frac{1}{2}\right)\frac{\pi x}{l}$

$\|X_n\|^2=\frac{l}{2}$

$\lambda_n=\left(n+\frac{1}{2}\right)^2\left(\frac{\pi}{l}\right)^2,\ n=0,1,2\ldots$

代入PDE：$\sum_{n=1}^{\infty}(T_n''X_n-a^2T_nX_n'')=\sum_{n=1}^{\infty}f_nX_n$

$X_n''=-\lambda_nX_n$

$\Rightarrow\sum_n(T_n''+a^2\lambda_nT_n-f_n)X_n=0$

$\Rightarrow T_n''+a^2\lambda_nT_n-f_n\neq 0$ ← 从后续计算可知，以式齐次BC下成立，非齐次BC下不成立

$u_{xx}\neq\sum_n T_n(t)X_n''(x)$ ← 逐次求导未必合法

例: $f(x)=x=\sum_{n=1}^{\infty}\frac{2l}{n\pi}(-1)^{n+1}\sin\frac{n\pi x}{l}$

⇓对x求导

$1 \overset{?}{=} \sum_{n} 2(-1)^{n+1}\cos\frac{n\pi x}{l}$

⇓再求导

$0 \overset{?}{=} \sum_{n} 2(-1)^{n}\frac{n\pi}{l}\sin\frac{n\pi x}{l}$

关于逐项求导失败的原因，回忆上次课末尾讨论的 Fourier 系数随 n 衰减行为与 $f(x), f'(x), f''(x)$ --- 连续性的关系.

一个简单具体的 PDE 例子:

$$u_{tt}-a^2u_{xx}=0 \qquad 0<x<l \qquad t>0$$

$$u|_{x=0}=0 \qquad u|_{x=l}=A \Leftarrow \text{常数}$$

$$u|_{t=0}=A\frac{x}{l}, \quad u_t|_{t=0}=0$$



$t=0$: 杆均匀拉伸并静止

$$\Rightarrow u(x,t)=A\cdot\frac{x}{l}=\sum_{n=1}^{\infty}A\frac{2}{\pi}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{l}$$

代入PDE与BC, IC, 显然成立.

杆保持原样.

代入PDE逐项求导 $\not\Rightarrow$???

级数本身是正确解，只是不可逐项求导

安全做法：不作逐项求导

$$\frac{2}{l}\int_0^l (u_{tt}-a^2u_{xx})X_n\,dx=\frac{2}{l}\int_0^l f(x,t)X_n\,dx$$

① ② ③

f的傅里叶系数

③ $=f_n(t)$

对参数t的求导移至积分号外

$$①=\frac{2}{l}\int_0^l u_{tt}(x,t)X_n(x)\,dx=\frac{2}{l}\int_0^l \frac{\partial^2}{\partial t^2}\big(u(x,t)X_n(x)\big)\,dx$$

$$=\frac{d^2}{dt^2}\Big[\frac{2}{l}\int_0^l u(x,t)X_n(x)\,dx\Big]=T_n''(t)$$

$=T_n(t)$

区间 $\longleftrightarrow$ 边界

Green's formula

$$\int_0^l (u''v-uv'')\,dx=(u'v-uv')\Big|_0^l$$

$\int_0^l[(u'v)'-u'v']\,dx$ $\qquad$ $\int_0^l[(uv')'-u'v']\,dx$

$$②=-\frac{2}{l}a^2\int_0^l u_{xx}X_n\,dx$$

$$\int_0^l (u_{xx}X_n-X_n''u)\,dx=(u_xX_n-uX_n')\Big|_0^l$$

$=-\lambda_n X_n$

$$\Rightarrow -\frac{2}{l}a^2\int_0^l u_{xx}X_n\,dx-\frac{2}{l}a^2\int_0^l \lambda_nX_nu\,dx=-a^2\frac{2}{l}(u_xX_n-uX_n')\Big|_0^l$$

$=②$

$$\Rightarrow ②=a^2\lambda_nT_n(t)-a^2\frac{2}{l}(u_xX_n-uX_n')\Big|_0^l$$

$$①+②=③ \Rightarrow T_n''+\underbrace{a^2\lambda_n}_{=\omega_n^2=(n+\frac12)^2(\frac{\pi a}{l})^2}T_n-a^2\frac{2}{l}(u_xX_n-uX_n')\Big|_0^l=f_n$$

$$\Rightarrow T_n''+\omega_n^2T_n=f_n+a^2\frac{2}{l}(u_xX_n-uX_n')\Big|_0^l$$

$u(l),u_x(0)$未给定，但恰好它们与0相乘！

$$=f_n+a^2\frac{2}{l}\Big[\underbrace{u_x(l)}_{h(t)}\underbrace{X_n(l)}_{(-1)^n}-u(l)\underbrace{X_n'(l)}_{0}-u_x(0)\underbrace{X_n(0)}_{0}+u(0)\underbrace{X_n'(0)}_{(n+\frac12)\frac{\pi}{l}}\Big]$$

$\mu(t)$

$$=f_n+a^2\frac{2}{l}\Big[(-1)^nh(t)+(n+\frac12)\frac{\pi}{l}\mu(t)\Big]$$

$=\tilde f_n$，"边界力"

$$=f_n+\tilde f_n$$

边界条件与外力作用得到统一

$X_n=\sin(n+\frac12)\frac{\pi x}{l}\Rightarrow X_n(0)=0,\ X_n(l)=(-1)^n$

$X_n'=(n+\frac12)\frac{\pi}{l}\cos(n+\frac12)\frac{\pi x}{l}\Rightarrow X_n'(0)=(n+\frac12)\frac{\pi}{l},\ X_n'(l)=0$

$$\Rightarrow \boxed{T_n''(t)+\omega_n^2T_n(t)=f_n(t)+\tilde f_n(t)}$$

$$\Rightarrow T_n(t)=\int_0^t\frac{1}{\omega_n}\sin\omega_n(t-\tau)\big(f_n(\tau)+\tilde f_n(\tau)\big)\,d\tau$$

$$u(x,t)=\sum_{n=0}^{\infty}T_n(t)X_n(x)=\cdots$$

换成其他边界条件，例如$u|_{x=0}=\mu(t)$，$u|_{x=l}=v(t)$，可同理推导.

会发现$X_n(0),X_n(l),X_n'(0),X_n'(l)$中该为0的自动为0，严丝合缝.

与未给定的$u(0),u(l),u_x(0),u_x(l)$相乘者

物理推导 + 直观理解：

移动法

取 $\mu(t)=0,\ h(t)\neq 0,\ f=0$，

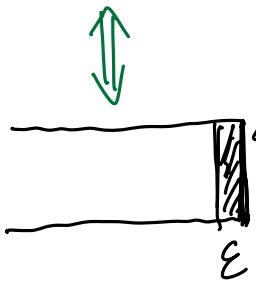
$$\Rightarrow \begin{cases} u_{tt}-a^2u_{xx}=0, & 0<x<l \\ u|_{x=0}=0, \quad u_x|_{x=l}=h(t) \end{cases}$$

$u_x|_{x=l}=h(t) \Rightarrow x=l$ 端单位面积外力 $F(t)=Eu_x|_{x=l}=Eh(t)$

总外力 $FS=ESh(t)$

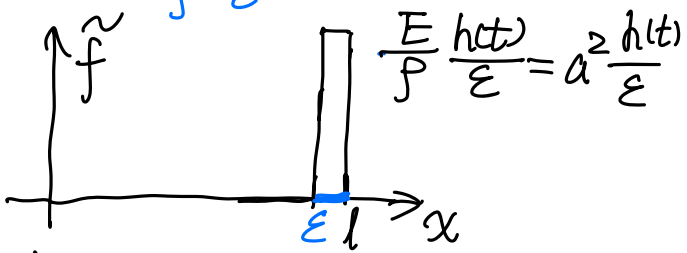


将右端 $x=l$ 单个截面上的外力分散至厚度为 ε 的薄层.



外力产生力速度 $\tilde{f}=\frac{\text{单位体积外力}}{\text{单位体积质量}}=\begin{cases} 0, & x<l-\varepsilon \\ \frac{E}{\rho}\frac{h(t)}{\varepsilon}=a^2\frac{h(t)}{\varepsilon}, & l-\varepsilon<x<l \end{cases}$

$$\begin{cases} u_{tt}-a^2u_{xx}=\tilde{f}(x,t) \\ u|_{x=0}=0, \quad u_x|_{x=l}=0 \end{cases}$$



齐次BC $\Rightarrow T_n''+a^2\lambda_nT_n=\tilde{f}_n(t)=\frac{2}{l}\int_0^l \tilde{f}(x,t)X_n(x)\,dx$

$\lambda_n=(n+\frac{1}{2})^2(\frac{\pi a}{l})^2$

$$=\frac{2}{l}\int_0^l \tilde{f}\sin(n+\tfrac{1}{2})\frac{\pi x}{l}\,dx=\frac{2}{l}\cdot\frac{a^2h(t)}{\varepsilon}(-1)^n=\frac{2}{l}a^2(-1)^nh(t)$$

$\sin(n+\frac{1}{2})\frac{\pi x}{l}\approx(-1)^n$，当 $x\approx l$

即 $T_n''+\omega_n^2T_n=\frac{2}{l}a^2(-1)^nh(t)$

与上页结果一致. $(-1)^n$ 也对上了

$$\Rightarrow T_n(t)=\int_0^t \frac{1}{\omega_n}\sin\omega_n(t-\tau)\tilde{f}_n(\tau)\,d\tau$$

$$=\frac{2}{l}a^2(-1)^n\int_0^t \frac{1}{(n+\frac{1}{2})\frac{\pi a}{l}}\sin(n+\tfrac{1}{2})\frac{\pi a}{l}(t-\tau)\cdot h(\tau)\,d\tau$$

$$\Rightarrow u(x,t)=\sum_{n=0}^{\infty}T_n(t)X_n(x)=\cdots$$

比较忙的同学可跳过本页

一个简单具体的PDE例子：

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \qquad 0<x<l \quad t>0 \\ u|_{x=0} = 0 \quad u|_{x=l} = A \Leftarrow \text{常数} \\ u|_{t=0} = A\frac{x}{l}, \quad u_t|_{t=0} = 0 \end{cases}$$



$t=0$：杆均匀拉伸并静止

$$\Rightarrow u(x,t) = A\frac{x}{l} = \sum_{n=1}^{\infty} A\frac{2}{\pi}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{l}$$

代入PDE与BC, IC, 显然成立.

杆保持原样.

代入PDE逐次求导 ⇒ ???

应用该方法, $u(x,t) = \sum_n T_n X_n = \sum_{n=1}^{\infty} T_n(t)\sin\frac{n\pi x}{l}$

$$T_n''(t) + \left(\frac{n\pi a}{l}\right)^2 T_n = a^2\frac{2}{l}\left[-u(l)X_n'(l)\right] = a^2\frac{2A}{l^2}n\pi(-1)^{n+1}$$

$$\begin{cases} T_n(0) = \frac{2A}{n\pi}(-1)^{n+1}, \quad T_n'(0) = 0 \end{cases}$$

边界力不可少！

$$\Rightarrow T_n(t) = \frac{2A}{\pi}\frac{(-1)^{n+1}}{n}$$

$$\Rightarrow u(x,t) = \sum_{n=1}^{\infty}\frac{2A}{\pi}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{l} = A\frac{x}{l}$$ ✓

由于 $\sin\frac{n\pi x}{l}\Big|_{x=l} = 0$，级数在 $x=l$ 处为0，但无妨。Fourier方法不在意孤立点上的函数值偏差. 在开区间 $(0,l)$ 上，级数都收敛至 $A\frac{x}{l}$，足够.

减去法 { BC齐次化 / BC+PDE同时齐次化 }　　卸方法

BC齐次化

1) BC齐次化

例: $u_{tt}-a^2u_{xx}=f,\ 0<x<l,$　$u|_{x=0}=\mu(t),\ u|_{x=l}=\nu(t)$

$u|_{t=0}=\phi(x),\ \frac{\partial u}{\partial t}|_{t=0}=\psi(x)$

$$u(x,t)=p(x,t)+w(x,t)$$

选p满足 $p|_{x=0}=\mu(t),\ p|_{x=l}=\nu(t)$　p不必满足PDE

$\Rightarrow w|_{x=0}=u|_{x=0}-p|_{x=0}=0,\quad w|_{x=l}=0$

减去p后，w满足齐次BC.

2) 求w.　$w_{tt}-a^2w_{xx}=(u-p)_{tt}-a^2(u-p)_{xx}$

$=u_{tt}-a^2u_{xx}-(p_{tt}-a^2p_{xx})$

$=f-(p_{tt}-a^2p_{xx})$　类似惯性力，因为w是以p为参照物看到的位移

$w|_{x=0}=w|_{x=l}=0$，齐次BC

$w|_{t=0}=u|_{t=0}-p|_{t=0}=\phi(x)-p|_{t=0}$

$\frac{\partial w}{\partial t}|_{t=0}=\frac{\partial u}{\partial t}|_{t=0}-\frac{\partial p}{\partial t}|_{t=0}=\psi(x)-\frac{\partial p}{\partial t}|_{t=0}$

弦相对于$p(x,t)$的初位移和速度

p的选取自由度较大.

① $p(x,t)=A(t)x+B(t)=(1-\frac{x}{l})\mu(t)+\frac{x}{l}\nu(t)$

①不能用于第二类BC $u_x|_{x=0}=\dots,\ u_x|_{x=l}=\dots$，因$p_x=A(t)$.

② $p(x,t)=A(t)(l-x)^2+B(t)x^2=\frac{\mu(t)}{l^2}(l-x)^2+\frac{\nu(t)}{l^2}x^2$

③ $A(t)x^2+B(t)x$

④ $A(t)x^2+B(t)$

⑤ ……

第三类BC. $(\alpha_1 u+\beta_1 u_x)_{x=0}=\mu(t)$, $(\alpha_2 u+\beta_2 u_x)_{x=l}=\nu(t)$

取P使得$(\alpha_1 P+\beta_1 P_x)_{x=0}=\mu(t)$, $(\alpha_2 P+\beta_2 P_x)_{x=l}=\nu(t)$

$P=A(t)x+B(t) \Rightarrow \begin{pmatrix}\beta_1, & \alpha_1\\ \alpha_2 l+\beta_2, & \alpha_2\end{pmatrix}\begin{pmatrix}A(t)\\ B(t)\end{pmatrix}=\begin{pmatrix}\mu(t)\\ \nu(t)\end{pmatrix} \Rightarrow \begin{pmatrix}A(t)\\ B(t)\end{pmatrix}=\cdots$

例: $u_{tt}-a^2u_{xx}=0$, $0<x<l$, $t>0$ $\quad u|_{x=0}=\mu(t)=0$

$u|_{t=0}=u_t|_{t=0}=0$ $\quad u|_{x=l}=\nu(t)=A\sin\omega t$

$P(x,t)=(1-\frac{x}{l})\mu+\frac{x}{l}\nu(t)=\frac{Ax}{l}\sin\omega t$,

$w_{tt}-a^2w_{xx}=-(P_{tt}-a^2P_{xx})=\frac{A\omega^2}{l}x\sin\omega t$

$w|_{x=0}=w|_{x=l}=0$, $\quad w|_{t=0}=-P|_{t=0}=0$

$w_t|_{t=0}=-P_t|_{t=0}=-\frac{A\omega}{l}x$

$w=w_1+w_2$ (初始条件, 非齐次项)

① $(w_1)_{tt}-a^2(w_1)_{xx}=0$

$w_1|_{x=0}=w_1|_{x=l}=0$, $\quad w_1|_{t=0}$, $\frac{\partial w_1}{\partial t}|_{t=0}=-\frac{A\omega}{l}x$

② $(w_2)_{tt}-a^2(w_2)_{xx}=-(P_{tt}-a^2P_{xx})$

$=\frac{A\omega^2}{l}x\sin\omega t$ $\quad g(x)$

解①: $w_1=\sum_{n=1}^{\infty}C_n\sin\omega_n t\sin\frac{n\pi x}{l}=\sum_{n=1}^{\infty}(-1)^n\frac{2A\omega}{n\pi\omega_n}\sin\omega_n t\sin\frac{n\pi x}{l}$

$\omega_n=\frac{n\pi a}{l}$

$\frac{\partial w_1}{\partial t}|_{t=0}=\sum_{n=1}^{\infty}C_n\omega_n\sin\frac{n\pi x}{l}=-\frac{A\omega}{l}x \Rightarrow C_n=(-1)^n\frac{2A\omega}{n\pi\omega_n}$

解②: $w_2 = \sum_{n=1}^{\infty} \frac{g_n}{\omega_n} \frac{\omega \sin\omega_n t - \omega_n \sin\omega t}{\omega^2 - \omega_n^2} \sin\frac{n\pi x}{l}$ （上次课结果）

$g_n = \frac{2}{l}\int_0^l g(x)\sin\frac{n\pi x}{l}dx = \frac{2A\omega^2}{n\pi}(-1)^{n+1}$

$\int_0^l x\sin\frac{n\pi x}{l}dx = ?$ 不用分部积分的小技巧 → $\int_0^l \cos\lambda x\,dx = \frac{\sin\lambda l}{\lambda}$

⇓对λ求导

$-\int_0^l x\sin\lambda x\,dx = \frac{l\cos\lambda l}{\lambda} - \frac{\sin\lambda l}{\lambda^2}$

$\lambda = \frac{n\pi}{l}$ ⇐ $\int_0^l x\sin\frac{n\pi x}{l}dx = \frac{l^2}{n\pi}(-1)^{n+1}$

$\int_0^\infty t^n e^{-t}dt = ?$ $\int_0^\infty e^{-\lambda t}dt = \lambda^{-1}$ n次求导 $\int_0^\infty (-t)^n e^{-\lambda t} = (-1)(-2)\cdots(-n)\lambda^{-(n+1)}$

⇓λ=1

$\int_0^\infty t^n e^{-t}dt = n!$

⇒ $w = w_1 + w_2$,

$u = p + w = p + w_1 + w_2$

$p = \frac{Ax}{l}\sin\omega t = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{2A}{n\pi}\sin\omega t\sin\frac{n\pi x}{l}$

$= \sum_{n=1}^{\infty}(-1)^n \frac{2A\alpha}{l}\frac{\omega_n\sin\omega t - \omega\sin\omega_n t}{\omega^2-\omega_n^2}\sin\frac{n\pi x}{l}$

$\omega_n = \frac{n\pi a}{l}$

BC + PDE 同时齐次化（特解齐次化）

PDE同上例：

$u_{tt} - a^2 u_{xx} = 0, \quad 0<x<l, \ t>0$

$u|_{x=0} = \mu(t) = 0$

$u|_{x=l} = v(t) = A\sin\omega t$

$u|_{t=0} = u_t|_{t=0} = 0$

$u = p + w, \quad p|_{x=0} = 0, \quad p|_{x=l} = A\sin\omega t$

p不仅满足BC，也满足PDE.

p也满足PDE

$p_{tt} - a^2 p_{xx} = 0$

$\Rightarrow w_{tt} - a^2 w_{xx} = -(p_{tt} - a^2 p_{xx}) = 0$

设 $p(x,t) = f(x)\sin\omega t, \quad f(0)=0, \quad f(l) = A$

$[\omega^2 f(x) + a^2 f''(x)]\sin\omega t = 0 \Rightarrow f(x) = A\dfrac{\sin\frac{\omega x}{a}}{\sin\frac{\omega l}{a}}$

（$\omega^2 f(x) + a^2 f''(x) = 0$）

$p = A\dfrac{\sin\frac{\omega x}{a}}{\sin\frac{\omega l}{a}}\sin\omega t$

$w_{tt} - a^2 w_{tt} = 0, \quad w|_{x=0} = w|_{x=l} = 0, \quad w|_{t=0} = -p|_{t=0} = 0$

$w_t|_{t=0} = -p_t|_{t=0} = -A\omega\dfrac{\sin\frac{\omega x}{a}}{\sin\frac{\omega l}{a}}$

$\Rightarrow w = \sum_{n=1}^{\infty}(-1)^{n+1}\dfrac{2Aa\omega}{l(\omega^2-\omega_n^2)}\sin\omega_n t\sin\dfrac{n\pi x}{l}$

$\int_0^l \sin\dfrac{\omega x}{a}\sin\dfrac{n\pi x}{l}dx = (-1)^n a\sin\dfrac{\omega l}{a}\dfrac{\omega_n}{\omega^2-\omega_n^2}$

展开 $p(x,t) = \sum_{n=1}^{\infty}(-1)^n\dfrac{2Aa\omega_n}{l}\dfrac{1}{\omega^2-\omega_n^2}\sin\dfrac{n\pi x}{l}\sin\omega t$

$u = p + w = \sum_{n=1}^{\infty}(-1)^n\dfrac{2Aa}{l(\omega^2-\omega_n^2)}(\omega_n\sin\omega t - \omega\sin\omega_n t)\sin\dfrac{n\pi x}{l}$

与只齐次化BC结果一致.

扩散方程 $u_t - D u_{xx} = 0 \quad 0<x<l \quad t>0$

$u|_{x=0} = A\sin\omega t, \quad u|_{x=l} = 0, \quad u|_{t=0} = 0$

齐次化BC.

$p(x,t) = A(1-\frac{x}{l})\sin\omega t$

$w_t - D w_{xx} = -(p_t - D p_{xx}) = -A\omega(1-\frac{x}{l})\cos\omega t = f(x,t)$

$w|_{x=0} = w|_{x=l} = 0, \quad w|_{t=0} = -p|_{t=0} = 0$

齐次化原理 $w(x,t) = \int_0^t v(x,t;\tau)d\tau, \quad v|_{t=\tau} = f(x,\tau) = \sum_{n=1}^{\infty} f_n(\tau)\sin\frac{n\pi x}{l}$

$v = \sum_{n=1}^{\infty} f_n(\tau) e^{-D(\frac{n\pi}{l})^2(t-\tau)} \sin\frac{n\pi x}{l}$

$\Rightarrow w(x,t) = \int_0^t v(x,t;\tau)d\tau = \sum_{n=1}^{\infty}\left[\int_0^t f_n(\tau) e^{-D(\frac{n\pi}{l})^2(t-\tau)} d\tau\right]\sin\frac{n\pi x}{l}$

$f_n(\tau) = \frac{2}{l}\int_0^l f(x,\tau)\sin\frac{n\pi x}{l}dx = -\frac{2A\omega}{n\pi}\cos\omega\tau$

$\alpha = D(\frac{n\pi}{l})^2$

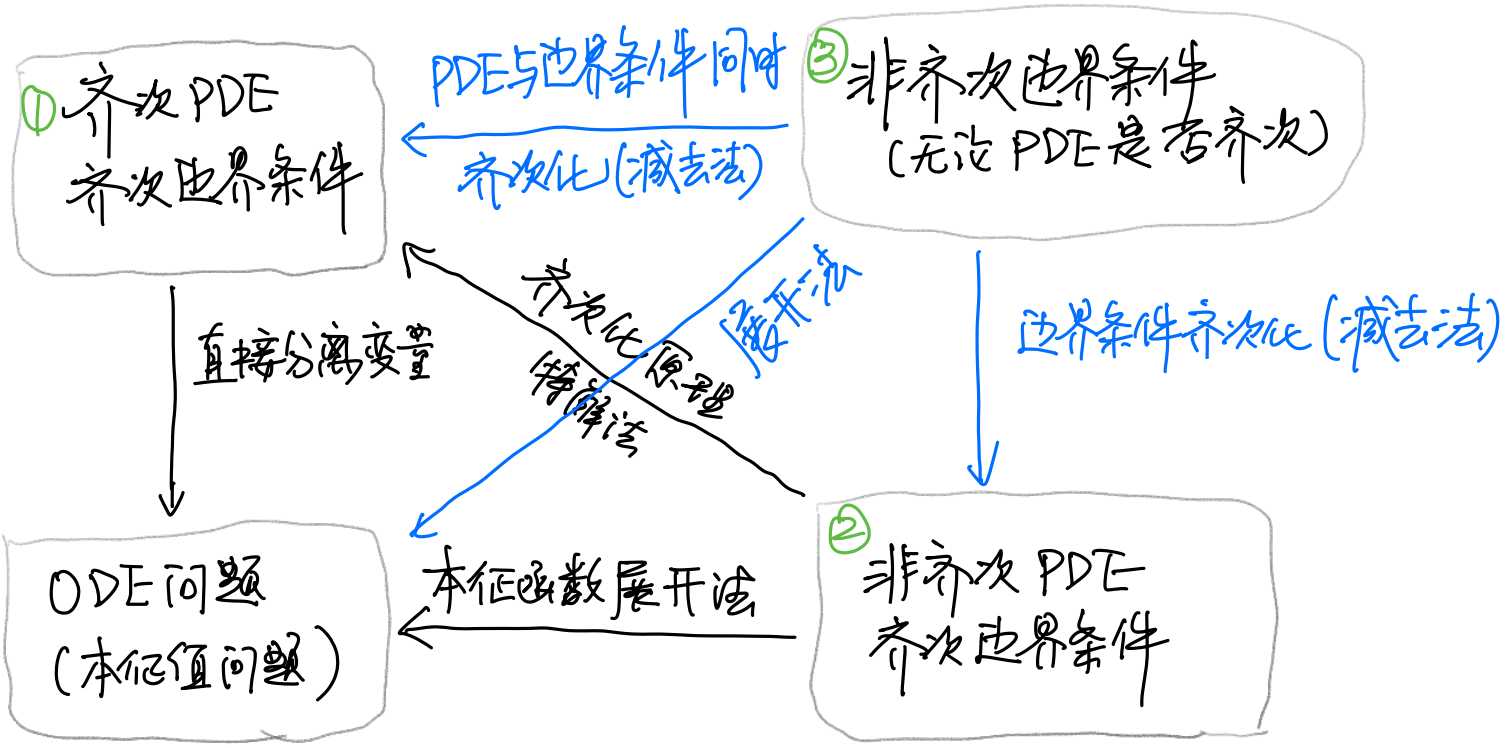
$\int_0^t e^{-\alpha(t-\tau)}\cos\omega\tau\, d\tau = e^{-\alpha t}\int_0^t e^{\alpha\tau}\frac{e^{i\omega\tau}+e^{-i\omega\tau}}{2} d\tau$

$= \frac{1}{\alpha^2+\omega^2}[\alpha(\cos\omega t - e^{-\alpha t}) + \omega\sin\omega t]$

$\Rightarrow w(x,t) = \sum_{n=1}^{\infty}\frac{2A\omega l^2}{D^2(n\pi)^4+\omega^2 l^4}\frac{1}{n\pi}[D(n\pi)^2 e^{-D(\frac{n\pi}{l})^2 t} - D(n\pi)^2\cos\omega t - \omega l^2\sin\omega t]\cdot\sin\frac{n\pi x}{l}$

$u = p + w = \ldots$

分离变量法


①齐次PDE
齐次边界条件
PDE与边界条件同时
齐次化(减去法)
③非齐次边界条件
(无论PDE是否齐次)
齐次化原理
特解法
降维法
直接分离变量
边界条件齐次化(减去法)
ODE问题
(本征值问题)
本征函数展开法
②非齐次PDE
齐次边界条件

ill-posed. 稳定性不成立的一个例子

$-\infty<x<\infty$, $y>0$

$u_{xx}+u_{yy}=0$

$u|_{y=0}=\phi(x)$, $\frac{\partial u}{\partial y}\Big|_{y=0}=\psi(x)$

① $\phi(x)=\psi(x)=0 \Rightarrow u=0$

② $\phi(x)=0$, $\frac{\partial u}{\partial y}\Big|_{y=0}=\psi(x)=e^{-\sqrt{n}}\sin(nx) \xrightarrow{n\to\infty} 0$

$n\to\infty$

$u_n(x,y)=\frac{1}{n}e^{-\sqrt{n}}\sin(nx)\sinh(ny)$, $\sinh(x)=\frac{e^x-e^{-x}}{2}$

$n\to\infty$. 对任意 $y>0$: $u_n(x,y)\xrightarrow{n\to\infty}\infty$

定解条件的微小改变 → 解的巨大改变 ⇒ 不稳定

BC从①变成②

当 $n\to\infty$ 时, BC ①和②之差 $\to 0$